

University of Connecticut
Computer Science & Engineering

Probabilistic Graphical Models (CSE 5830)
Fall 2026

Instructor

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Course Description

Probabilistic graphical models (PGMs) provide a powerful and flexible framework for reasoning under uncertainty and analyzing large-scale, complex, and heterogeneous data. Building on foundations in probability theory, graph theory, statistics, and machine learning, PGMs unify principles of representation, inference, and learning. They underpin state-of-the-art methods across diverse domains, including natural language processing, computer vision, healthcare and biomedical informatics, social and information networks, autonomous systems, and scientific discovery.

This course introduces the core principles of PGMs, including Bayesian networks, Markov random fields, and factor graphs. Students will learn exact and approximate inference algorithms, as well as parameter and structure learning techniques. The course will also explore modern extensions and connections to deep learning, uncertainty analysis, variational methods, and generative models, highlighting the role of PGMs in contemporary AI research and applications.

Prerequisites

Familiarity with programming, basic linear algebra (matrices, vectors, matrix-vector multiplication), basic probability (random variables, distributions, densities, marginalization), basic statistics (moments, typical distributions, regression), and algorithms (dynamic programming, basic data structures, complexity) is assumed.

It is helpful to have some experience programming for machine learning (e.g. NB, HMM) and ability to deal with abstract mathematical concepts.

Texts and Materials

The required textbook for this course is:

Probabilistic Graphical Models: Principles and Techniques, D. Koller & N. Friedman, MIT Press, 2009 (ISBN: 9780262013192).

<https://mitpress.mit.edu/books/probabilistic-graphical-models>

Additional textbooks are:

- *Bayesian Reasoning and Machine Learning*, David Barber, Cambridge University Press 2012. Free online.
<http://web4.cs.ucl.ac.uk/staff/D.Barber/pmwiki/pmwiki.php?n=Brml.HomePage>
- *Probabilistic Graphical Models*, Luis Enrique Sucar, Springer, 2021
<https://link.springer.com/book/10.1007/978-3-030-61943-5>

- *An Introduction to Probabilistic Graphical Models*, Michael I. Jordan, in preparation. Copies of certain chapters will be made available.
<https://people.eecs.berkeley.edu/~jordan/prelims/>
- *Deep Generative Models*, Jakub M. Tomczak, Springer Nature, 2024.
<https://link.springer.com/book/10.1007/978-3-031-64087-2>

Lectures

Lecture: Monday & Wednesday, 2:30 PM -3:45 PM, CHM T309.

The conceptual and theoretical course contents will be delivered primarily in the lectures, complemented by readings from the textbook. You should review readings prior to attending a lecture. Some of the topics may seem difficult at first. It is therefore absolutely essential that you ask questions whenever something is said which you do not understand.

You are expected to attend all lectures. If you are out of class for an extended period of time because of sickness, notify your instructor as soon as possible, and see your instructor immediately upon your return in order to determine how to catch up. If you have missed a significant portion of the semester due to illness, it is recommended that you resign from the course.

Outcome Measures

Class participation. Active participation through regular attendance and engagement in class discussions is an essential component of this course. To encourage active learning, there will be occasional short pop quizzes, typically consisting of a single question, based on recent lecture material.

Homework. There will be four or five homework assignments.

Midterm. We will have an in-class exam on Wednesday October 21th.

Advanced Topic Presentation. Each student is required to select a paper from the provided list of papers on advanced topics in PGM, published in leading journals or conferences, and present it to the class. The selected topic may be related to your final project and can also serve as one of its references.

- **Presentation Schedule.** Tentatively set for November 9, 11, 16, and 18.
- **Format.** Each presentation should be 30 minutes, followed by 5 minutes of Q&A. Slides are required.
- **Evaluation.** Presentations will be assessed based on quality, clarity, and depth of understanding. Peer feedback will contribute to the evaluation.
- **Preparation.** Students must select the paper by Wednesday November 4th.

Course Project. This is a project-based course, and there will be no final exam. Instead, students will work on a course project throughout the semester.

The project must focus on a well-defined problem in probabilistic graphical models (PGMs) or probabilistic deep learning. The project proposal should clearly describe the probabilistic model being used and the inference problem being addressed. Projects may be theoretical or applied.

The project will have 4 deliverables:

- 1- Proposal presentation, Wednesday October 26th
- 2- Proposal report: 2 pages, Due: Friday October 28th
- 3- Final Presentation: in class presentation of your project on December 7th and 9th.
- 4- Final Report: minimum 5 pages, In the format of IEEE conference papers, Due: Wednesday December 16th.

- **Team Formation.** You are responsible for forming project teams of 2 people by Friday, October 9th. Once you have formed your group, please send one email per team to the class instructor, list with the names of all team members. If you have trouble forming a group, please send the class instructor to help you to find project partners.
- **Project Proposal.** Each team must submit a brief project proposal that includes an overview of the project idea and a short survey of related work on the topic. In addition, teams are required to meet with the instructor before the proposal deadline to discuss their ideas and receive feedback. This meeting will help ensure the project is well aligned with the course objectives.

Proposals should be approximately two pages long, and should include the following information:

- Project title and list of group members.
- Overview of project idea. This should be approximately half a page long.
- A short literature survey of 4 or more relevant papers. The literature review should take up approximately one page.
- Description of potential data sets to use for the experiments.
- Plan of activities, and how you plan to divide up the work.

The project proposal will be due on October 28th and should be submitted electronically via HuskyCT.

Note that even though you can use data sets you have used before, you cannot use work that you started prior to this class as your project.

You also need to give a presentation on your proposal (10 minutes) on Monday October 26th and consider the feedbacks from the class in your proposal.

The grading breakdown for the proposal is as follows:

35% for clear and concise description of proposed method

35% for literature survey that covers at least 5 relevant papers

10% for plan of activities

10% for quality of writing

10% for quality of presentation

- **Final Report.** Your final report is expected to be 5 pages, in the format of the IEEE conference papers. It should have roughly the following format:
 - Introduction: problem definition and motivation
 - Background & Related Work: background info and literature survey
 - Methods: overview of your proposed method, Intuition on why should it be better than the state of the art, and details of models and algorithms that you developed
 - Results: Description of your experiments and results and a list of questions your experiments are designed to answer
 - Conclusion: discussion and future work

The grading breakdown for the final report is as follows:

10% for introduction and literature survey

30% for proposed method (soundness and originality)

30% for correctness, completeness, and difficulty of experiments and figures

10% for empirical and theoretical analysis of results and methods

20% for quality of writing (clarity, organization, flow, etc.)

The final report will be due on Wednesday, December 16th, and should be submitted via HuskyCT.

- **Presentation.** All project teams will present their work at the end of the semester on December 7th and 9th. Each team will be given a timeslot during which they will give a slide presentation to the class, similar in style to a conference presentation. If applicable, live demonstrations of your software are highly encouraged.

Course outline

- Introduction and Overview
- Review of Probability Theory
- Directed Graphical Model Representation
- Undirected Graphical Model Representation
- Local Conditional Probability Distributions
- Template-Based Graphical Models
- Exact Inference by Variable Elimination
- Clique Trees
- Belief Update and Message Passing
- Approximate Inference as Optimization
- Approximate Inference by Sampling
- MAP Inference
- Probabilistic Deep Learning and Bayesian Neural Networks
- Variational Inference and Deep Latent-Variable Models
- Probabilistic Generative Models: VAEs, Autoregressive Models
- Diffusion and Score-Based Generative Models
- Parameter Estimation in Undirected Graphical Models
- Parameter Estimation in Directed Graphical Models with Missing Data
- Graphical Model Structure Learning

Grading Criteria

Class participation	10%
Homework	10%
Midterm	20%
Proposal	5%
Advanced topics study	15%
Final report	35%
Final presentation	5%

Final grade

A+	100 – 97
A	96 - 93
A-	92 - 90
B+	89 – 87
B	86 – 83
B-	82 - 80
C+	79 – 77
C	76 - 73
C-	72 - 70
D+	69 – 67
D	66 – 63
D-	62 - 60
F	59 or less

Correct use of AI

It is important that AI tools be used to enhance, rather than to replace learning. This means that AI tools should not be used to generate final output or solutions. Instead, AI tools should be used to seek clarification on difficult concepts, generate additional examples or practice problems, seek feedback on solution attempts, ask for short hints if you are stuck on a problem set question, or research new topics. When using generative AI tools, you should keep in mind that these tools can provide incorrect information although written in a style that inspires confidence. Ultimately, you are responsible for assessing the accuracy of any AI generated content.

It is important to understand that the purpose of problem sets and other assignments is to help you learn the material, deepen your understanding, and develop your problem-solving skills. Incorrect or excessive use of AI may help you to be quicker or do better on problem sets, creating an illusion of success, but will ultimately cause failure since you will not properly learn the course content, not be able to do well on the exams, and not be able to answer questions during your class presentation.

Acknowledging use of AI

Any use of AI for any problem set or other assignment must be properly documented and credited. For each such use, please include a brief (2-3 sentences) description of what specific tool was used, what prompt was used to generate material, and how that material was incorporated into the assignment.

Academic Integrity

We will follow the University Policy on Academic Integrity. For more information, see the related Student Conduct Rules. The URL of the UConn Student Conduct Rules web page is:

<http://community.uconn.edu/the-student-code-preamble/>

All academic work must be your own. Collaboration, usually evidenced by unjustifiable similarity in assignments, is never allowed. After an appropriate informal review, if any students are found in violation of maintaining academic integrity, sanctions will be imposed, which can be as severe as failing the course for reasons of academic dishonesty. Especially flagrant violations will be considered under formal review proceedings, which can call for harsher sanctions including expulsion from the University. If you ever have any questions or concerns regarding the policy, particularly as it relates to this course, please talk to your instructor.

Disability

If you have a diagnosed disability (physical, learning, or psychological) that will make it difficult for you to carry out the course work as outlined, or that requires accommodations such as recruiting note-takers, readers, or extended time on exams or assignments, please advise the instructor during the first two weeks of the course so that we may review possible arrangements for reasonable accommodations.

HuskyCT Site

Assignments are submitted via HuskyCT. Students are expected to visit the site on a regular basis for course materials and announcements.